

MANTHAN

NEET 2027

PHYSICAL CHEMISTRY

CHEMICAL KINETICS

→ LECTURE -01

Subtopic – RATE & RATE OF REACTION

Mole concept

One shot

Part-2 → Youtube

live → 6.00 pm

Chem → 8.30 am
Phy → 10.45 am

→ New chapter

→ Chemical Kinetics

→ Sunday → test = liquid solution
timing = only chemistry

Sarvesh Sir

Today's goal



- ▶ **RATE**
- ▶ **RATE OF REACTION**
- ▶ 3 Question
= 12 Marks



TYPE OF REACTION



Rate के Bases पर $\rightarrow$ Reaction को 3-type

① Very fast Rxn $\rightarrow$ does not measure of Rate of reaction

(2) Very slow Reaction $\rightarrow$ does not measure of Rate of rxn

(3) Moderate Reaction or Avg. speed से होती $\rightarrow$ measure of Rate of rxn
or slow Reaction

$\rightarrow$ minute

$\rightarrow$ hour



RATE

Change in Concentration of Reactant or product per unit time or per second.

$$\text{Rate} = \frac{\text{change in Concentration}}{\text{time taken}}$$



initial value $\rightarrow$ A_0

$0 \rightarrow t=0$

After 't' sec $\rightarrow$ $\text{A}_t = (\text{A}_0 - x)$
Con. of Reactant

$x \rightarrow t \text{ sec}$

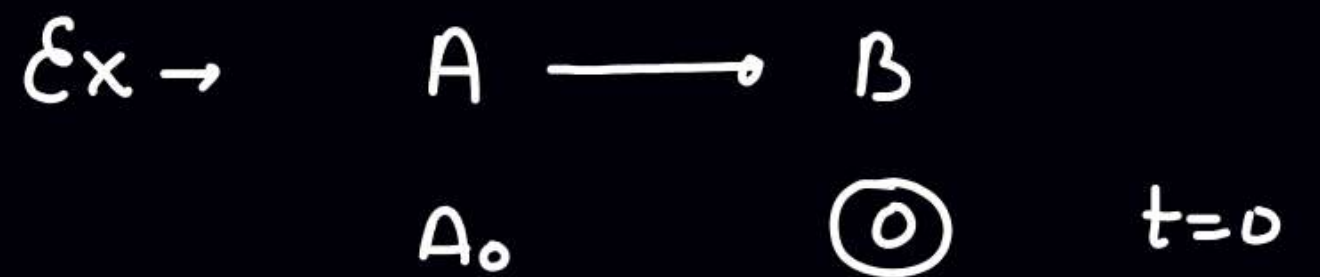
$$R_A = \frac{C_f - C_i}{t_f - t_i} = \frac{\Delta C}{\Delta t}$$

$$R_A = \frac{A_t - A_0}{t - 0} = \frac{A_t - A_0}{t}$$

$$R_A = \frac{\cancel{A_0} - x - \cancel{A_0}}{t} = -\frac{x}{t}$$

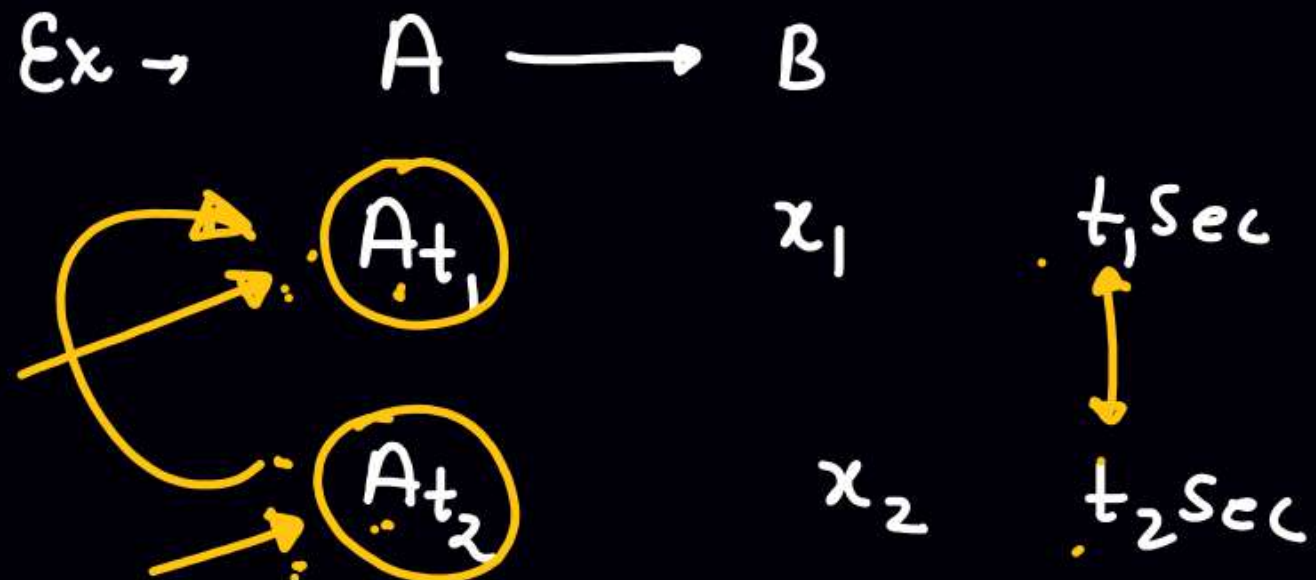
Represent
 $\rightarrow$ change in con.

$$R_B = \frac{C_f - C_i}{t_f - t_i} = \frac{x - 0}{t - 0} = \frac{x}{t}$$



$$R_A = \frac{A_t - A_0}{t - 0} = \frac{A_t - A_0}{t}$$

$$R_B = \frac{x - 0}{t - 0} = \frac{x}{t}$$



$$\text{Rate} = \frac{\Delta C}{\text{time taken}} = \frac{C_f - C_i}{t_f - t_i}$$

$$(R_{\text{rate}})_A = \frac{A_{t_2} - A_{t_1}}{t_2 - t_1}$$



Con. of Reactant 'A' initially = 2 mole/L
and After 10 Sec. Con. of A is
1.2 mole/litre. Calculate of Rate
of A

$$R_A = \frac{C_f - C_i}{\text{time taken}}$$

$$= \frac{1.2 - 2.0}{10}$$

$$R_A = -\frac{0.8}{10}$$

~~a) $-\frac{0.8}{10}$~~

b) $+\frac{0.8}{10}$

c) $-\frac{1.2}{10}$

d) $+\frac{1.2}{10}$



After 10 Sec and 15 Sec, Con. of
Reactant A is 1.2 mole/litre &
0.8 mole/litre respectively.

Calculate Rate of A



$A_0 = 1.2$

$A_t = 0.8$

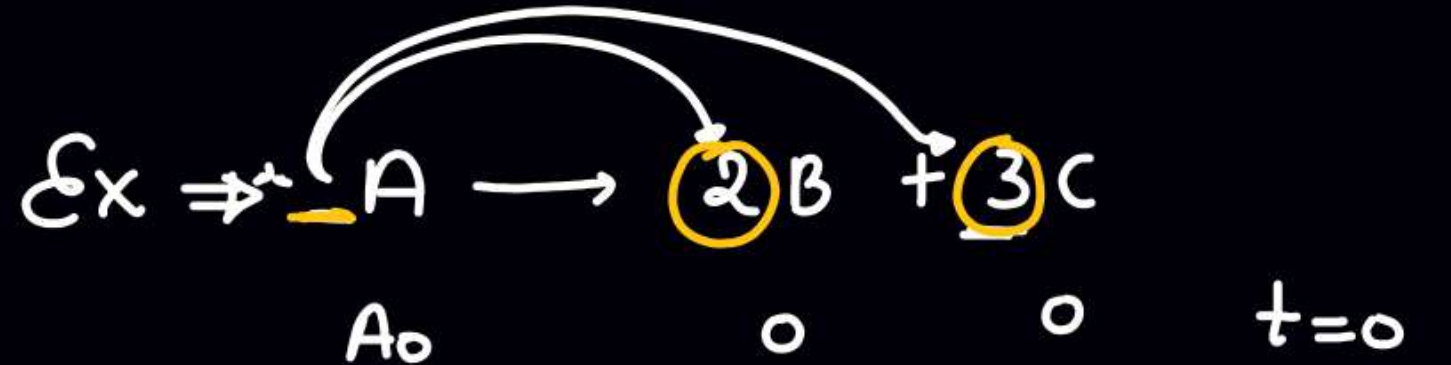
a) $\frac{0.4}{5}$

~~b) $-\frac{0.4}{5}$~~

c) $-\frac{0.8}{5}$

d) $-\frac{1.2}{5}$

$$R_A = \frac{0.8 - 1.2}{15 - 10} = -\frac{0.4}{5}$$



$$R_A = \frac{A_t - A_0}{t} = \frac{\text{A}_0 - x - \text{A}_0}{t} = -\frac{x}{t}$$

$$R_B = \frac{2x - 0}{t} = \frac{2x}{t} = +2\left(\frac{x}{t}\right)$$

$$R_C = \frac{3x - 0}{t} = \frac{3x}{t} = +3\left(\frac{x}{t}\right)$$

$$\rightarrow r_A \neq r_B \neq r_C$$

R_A = A किस Rate से dissociate हो रहा है।

R_A = Rate of disappearance

R_B & R_C = B & C किस Rate से form हो रहे हैं।

or
 Rate of formation (Appearance)

R_A, R_B, R_C = individual Rate

Rate of Reaction

$$R.O.R. = \frac{\text{individual rate}}{\text{Stoichiometric coefficient}}$$

Reaction
Balance में
- आएगा

$$R.O.R. = -\frac{\gamma_A}{1} = +\frac{\gamma_B}{2} = +\frac{\gamma_C}{3}$$

$$\begin{aligned} &= -\frac{\frac{x}{t}}{1} = \frac{2x/t}{2} = \frac{3x/t}{3} \\ &= \frac{x}{t} = \frac{x}{t} = \frac{x}{t} \end{aligned}$$

Csd = Rate of rxn = always +ve होता है।

Csd →



$$R.O.R. = -\frac{\gamma_A}{\eta_1} = -\frac{\gamma_B}{\eta_2} = +\frac{\gamma_C}{\eta_3}$$

$$\begin{array}{l|l} \gamma_A = \frac{\Delta A}{t} & * -\frac{\gamma_A}{\eta_1} = \frac{\gamma_C}{\eta_3} \\ \gamma_B = \frac{\Delta B}{t} & * -\frac{\gamma_B}{\eta_2} = \frac{\gamma_C}{\eta_3} \\ \gamma_C = \frac{\Delta C}{t} & \end{array}$$

Q → क्या R.O.R., individual Rate से ज्यादा हो सकता है।
= ?

Solⁿ → yes



$$R.O.R. = -\frac{\gamma_A}{2} = +\frac{\gamma_B}{3}$$

$$R.O.R. < \gamma_A$$

$$R.O.R. < \gamma_B$$



$$R.O.R. = -\frac{\gamma_{N_2}}{1} = -\frac{\gamma_{O_2}}{\frac{1}{2}} = +\frac{\gamma_{N_2O}}{1}$$

$$R.O.R. = -\gamma_{N_2} = -2 \cdot \gamma_{O_2} = +\gamma_{N_2O}$$

$$R.O.R. = -2 \cdot \gamma_{O_2}$$

$$\rightarrow R.O.R. > \gamma_{O_2}$$

$$\underline{C_{sd}} \rightarrow R \cdot O \cdot R = \frac{R_{ind}}{\eta}$$

① $\eta = \text{stoichiometric coefficient}$

$$\eta = 1 \rightarrow R \cdot O \cdot R = R_{\text{individual rate}}$$

② $\eta > 1 \rightarrow R \cdot O \cdot R < R_{ind.}$

③ $\eta < 1 \rightarrow R \cdot O \cdot R > R_{ind.}$

CHEMICAL KINETICS

Qsd-SHEET-1

Q-1 Select Correct Representation of Rate of Reaction



a) $-\frac{r_A}{3} = +\frac{r_B}{2}$

b) $-2r_A = +3r_B$

c) $+\frac{r_B}{2} = -\frac{r_A}{3}$

d) All Correct

Q-2 Rate of reaction expressed by

$$-\frac{r_B}{2} = \frac{r_A}{3} = \frac{r_D}{1} = -\frac{r_C}{3}$$



Q-3



gf Rate of disappearance of A is $-1.5 \times 10^3 \text{ mole} \cdot \text{litre}^{-1} \cdot \text{Sec}^{-1}$

than Calculate rate of formation of C

a) -7.5×10^3

b) $+7.5 \times 10^3$

c) 1.5×10^3

d) 0.3×10^3



After 10 seconds the concentration of product B formed is 2.8 mole/litre. Calculate rate of disappearance of A

a) 0.28

b) 0.84

c) 0.093

d) 0.14



Select Correct representation of Rate relation.

a) $\ln r_A = 2 \ln r_B$

b) $\ln(-r_A) = \ln(r_B)$

c) $\ln(-r_A) = \ln B + \ln 2$

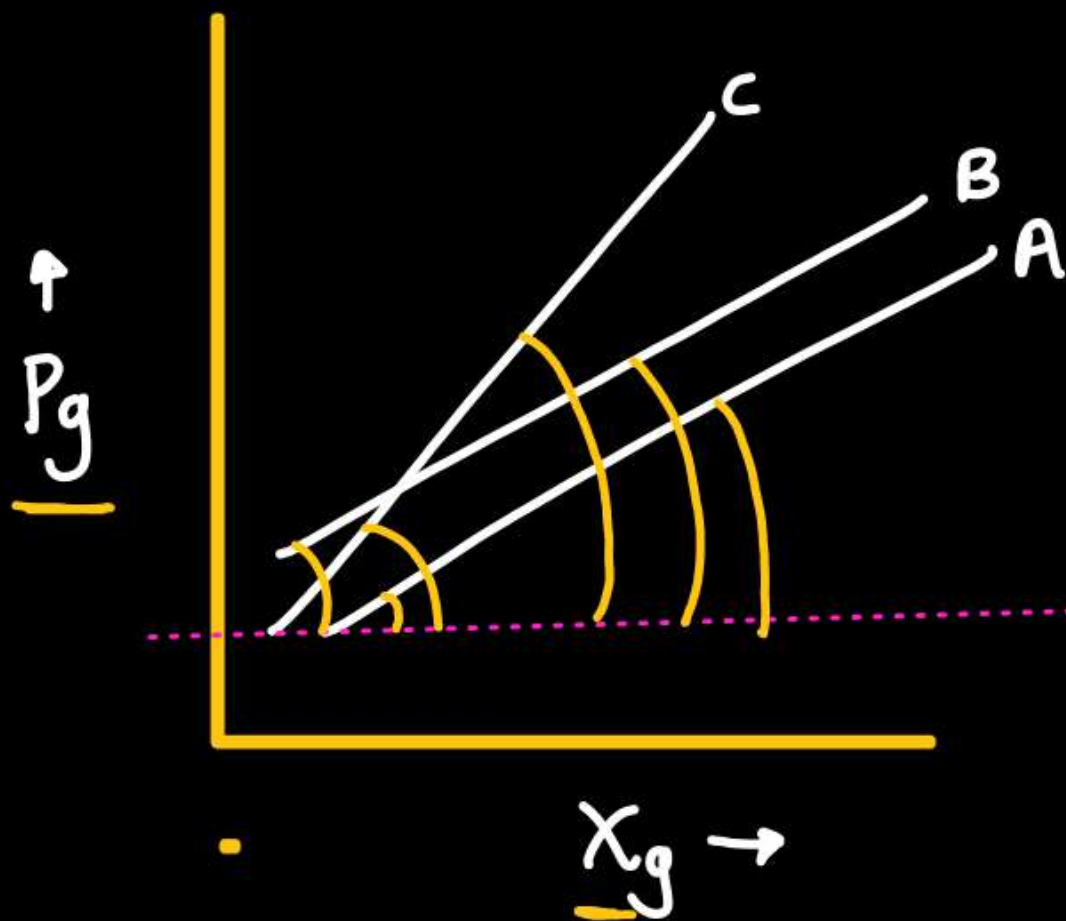
d) $\ln(-r_A) + \ln 2 = \ln B$

LIQUID SOLUTION

Qsd-SHEET-11

Q-1 Gas-A, B & C represent in graph

Select incorrect statements.



☒ a) Gas-C is more soluble in H_2O

☐ b) Value of K_H is lowest of gas-A

☐ c) Gas-B is less soluble than gas-A

☒ d) Solubility all equal because temp. is constant

(a, d)

$$(K_H)_C > (K_H)_B > (K_H)_A$$

$$\text{Solubility} \propto \frac{1}{K_H}$$

$$P_g = (K_H)_g \cdot X_g$$
$$\frac{P_g}{Y} = m \quad X_g = x$$

$$\rightarrow \text{Slope} = (K_H)_g$$

Q-2 which gas is more soluble in Benzene solvent

a) SO_2

b) $\text{CO}_2 \rightarrow \mu=0 \rightarrow$ dipole moment

c) $\text{CH}_4 \rightarrow$ nonpolar

d) NH_3

non polar - solvent

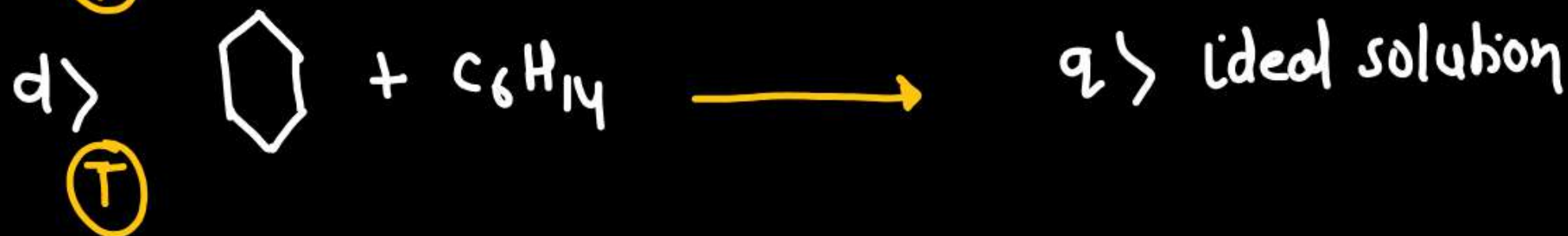
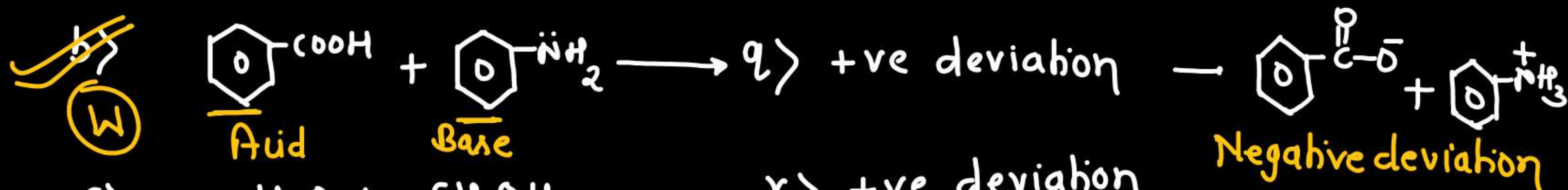
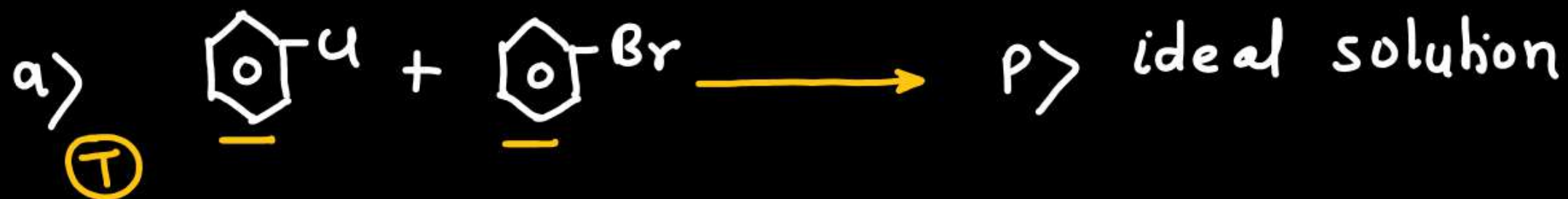
more soluble $\rightarrow$ non polar
solute (gas)

like dissolve like

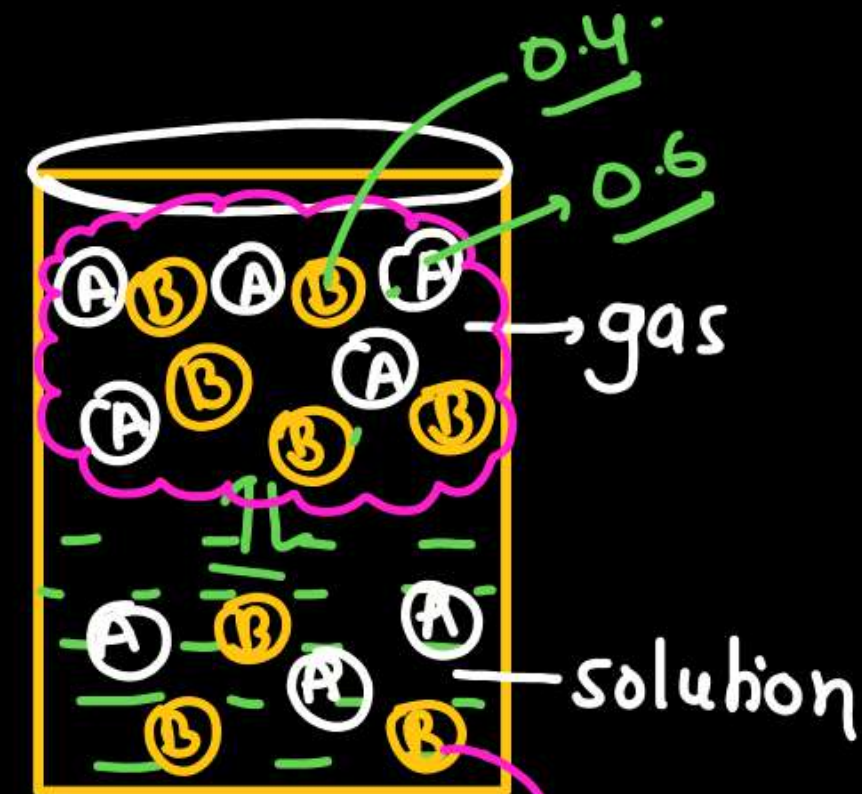
Q-3 Select incorrect match

List-I (solution)

List-II (type of solⁿ)



Q-4 If total pressure of gas mixture is 10 atm and mole fraction of gas-A in vapour phase is 0.6. Calculate mole of gas-B in solution. If 10 mole of H_2O solvent used in solution. $(K_H)_A = 1 \times 10^4 \text{ atm}$, $(K_H)_B = 1 \times 10^5 \text{ atm}$.



~~a) 4×10^4~~

b) 4×10^5

c) 6×10^4

d) 6×10^5

e) NONE

$n_B = \frac{4}{10^4}, n_B = 4 \times 10^{-4}$

$P_g = (K_H)_g \cdot X_g$

$P_B = (K_H)_B \cdot X_B$

$4 = 1 \times 10^5 \cdot \left(\frac{n_B}{n_B + 10} \right)$

$4 = 10^5 \times \frac{n_B}{10}$

vap. phase

$P_T = 10 \text{ atm} = P_A + P_B$

$Y_A = 0.6, Y_B = 0.4$

$P_B = Y_B \cdot P_T$
 $= 0.4 \times 10 = \underline{4 \text{ atm}}$

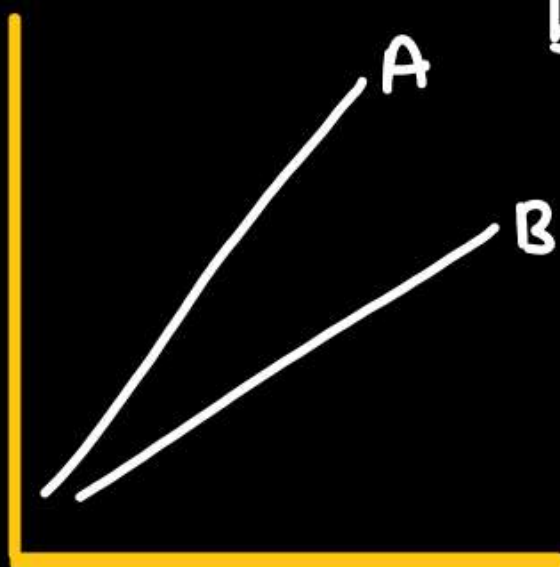
Q-5 Henry's Const. of gas-A is greater than gas-B

Select correct graph for partial pressure of gas Vs mole

fraction of water. $P_g = -(K_H)_{H_2O} \chi_{H_2O} + (K_H)_g P_g$ Vs χ_{H_2O}

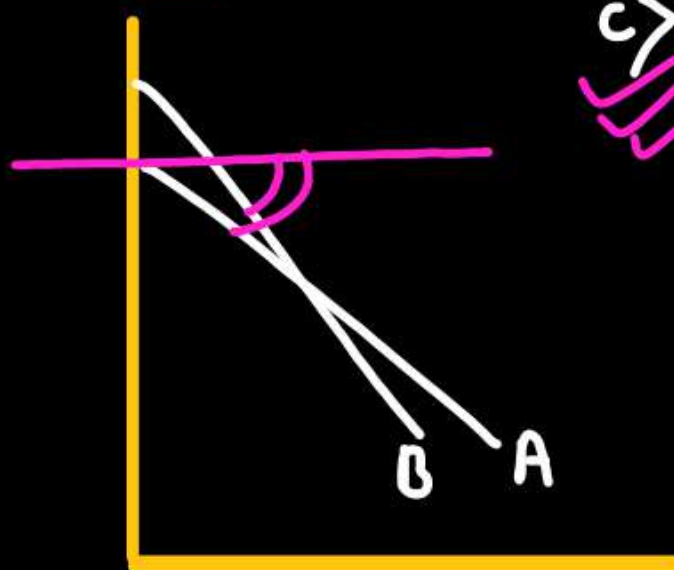
$$\frac{P_g}{y} = -\underline{m} x + c$$

a)

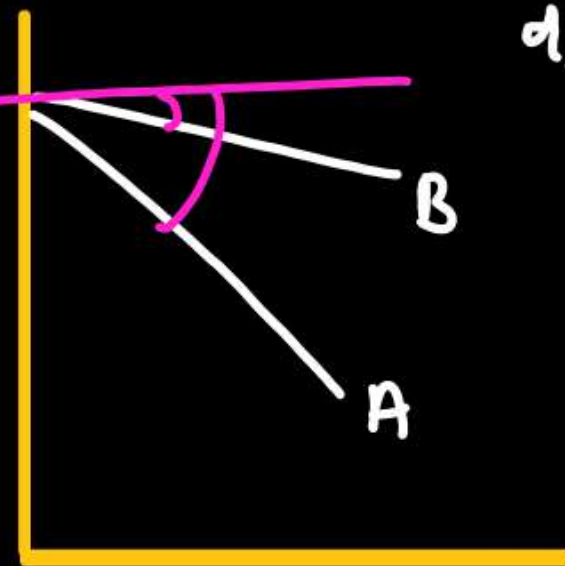


x

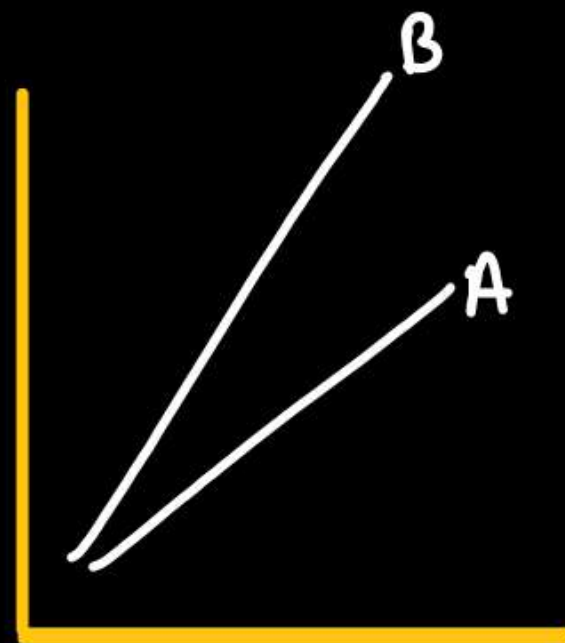
b)



c)



d) X



Slope = K_H $(K_H)_A > (K_H)_B$